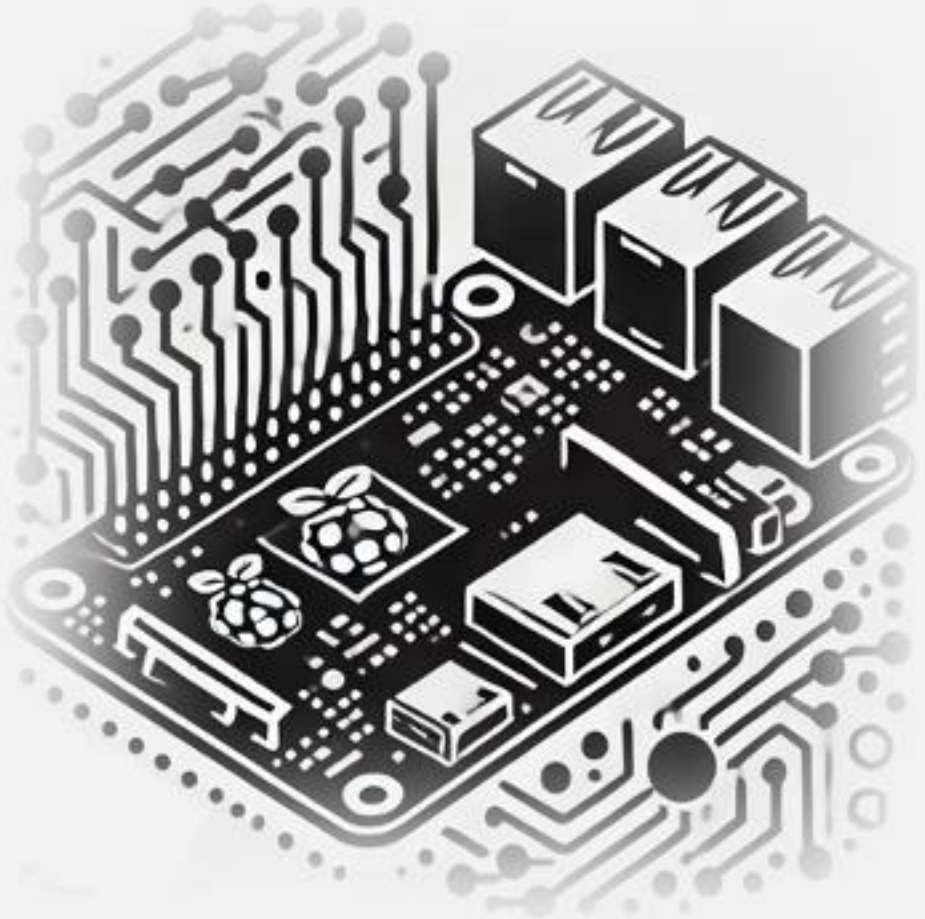
IESTI05 – Edge AI

Machine Learning System Engineering



5. Image Classification: Introduction to AI Edge LiteRT

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Computer Vision Recognition Tasks

Image Classification

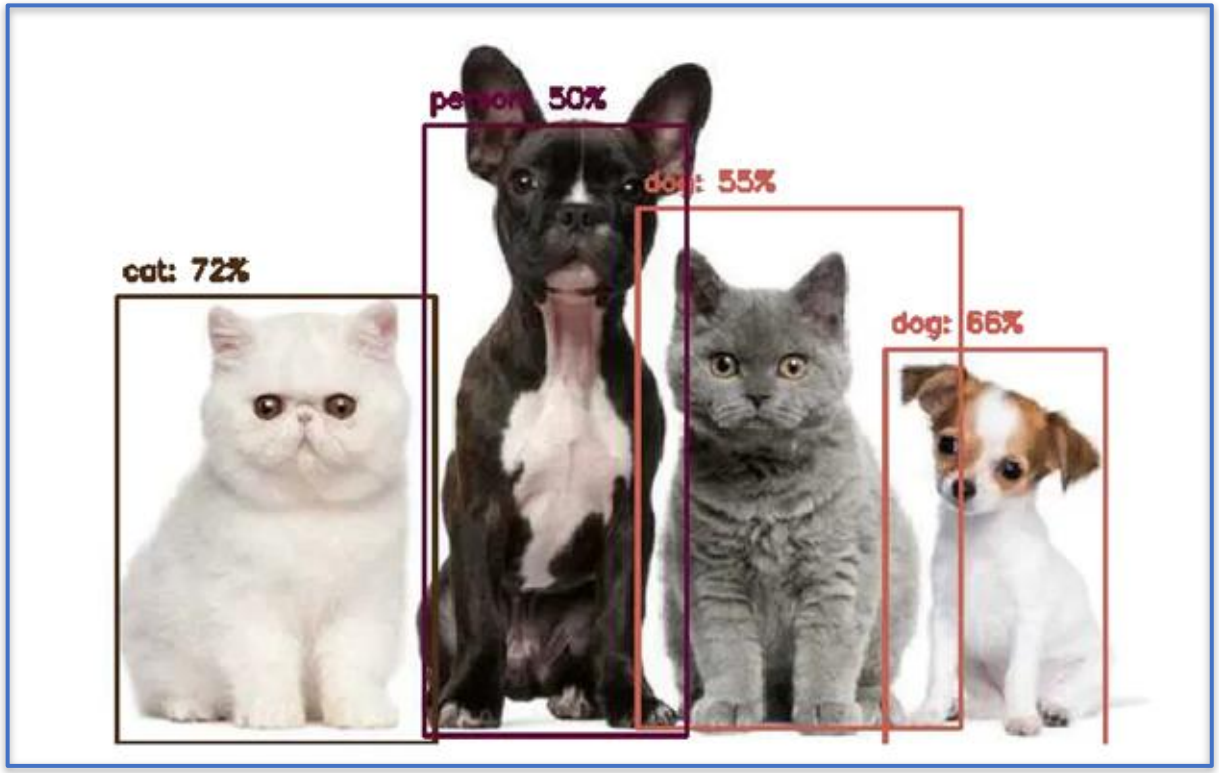
(**Multi-Class** Classification)

Cat: 70%

Dog: 80%

Object Detection

Multi-Label Classification + Object **Localization**



Computer Vision Recognition Tasks

Instance Segmentation

Each pixel in an image IS CLASSIFIED into a predefined category.

Pose Estimation

Key points (or landmarks) on the object, such as joints on a human body are detect

Image Classification

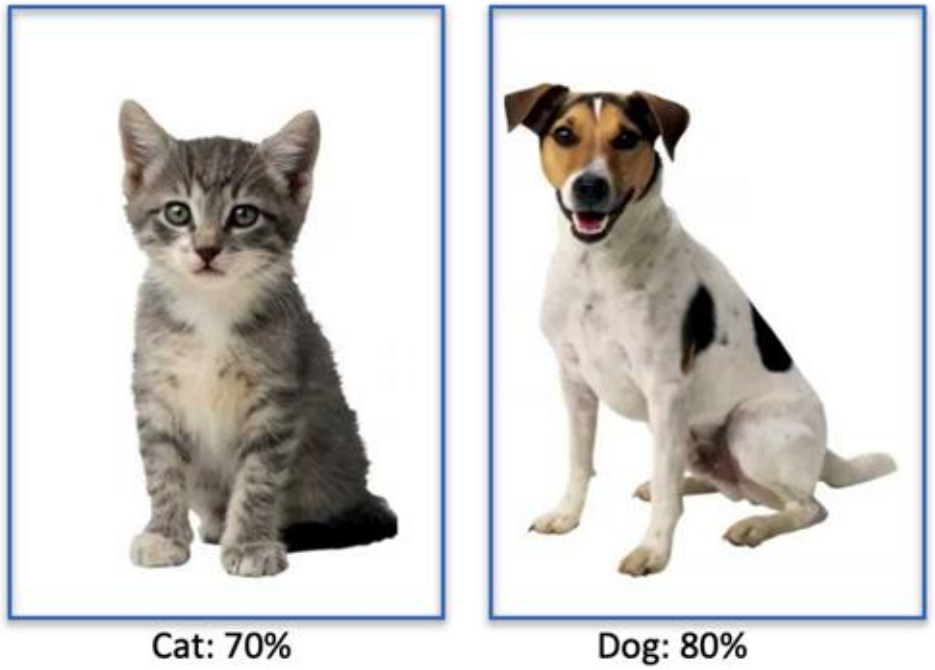
What is Image Classification?

Definition

Image classification is a fundamental task in computer vision that involves **categorizing an image into one of several predefined classes**.

Key Characteristics

- Assigns a single label to the entire image
- Based on visual content analysis
- Uses machine learning algorithms
- Mimics human visual perception



Real-World Applications

🏥 Healthcare

Medical image analysis, X-ray and MRI diagnostics

🌾 Agriculture

Crop health monitoring, disease detection

🚗 Automotive

Autonomous vehicles, road sign recognition

🔒 Security

Surveillance systems, threat detection

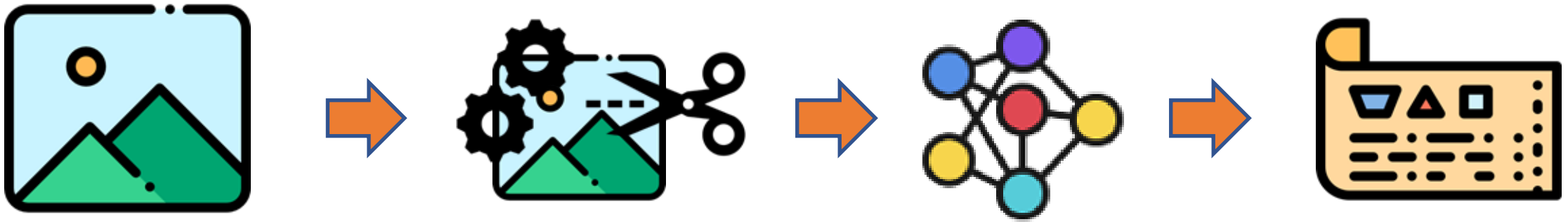
🛒 Retail

Visual search, inventory management

🌍 Environmental

Satellite imagery analysis, monitoring

Image Classification: Inference Pipeline



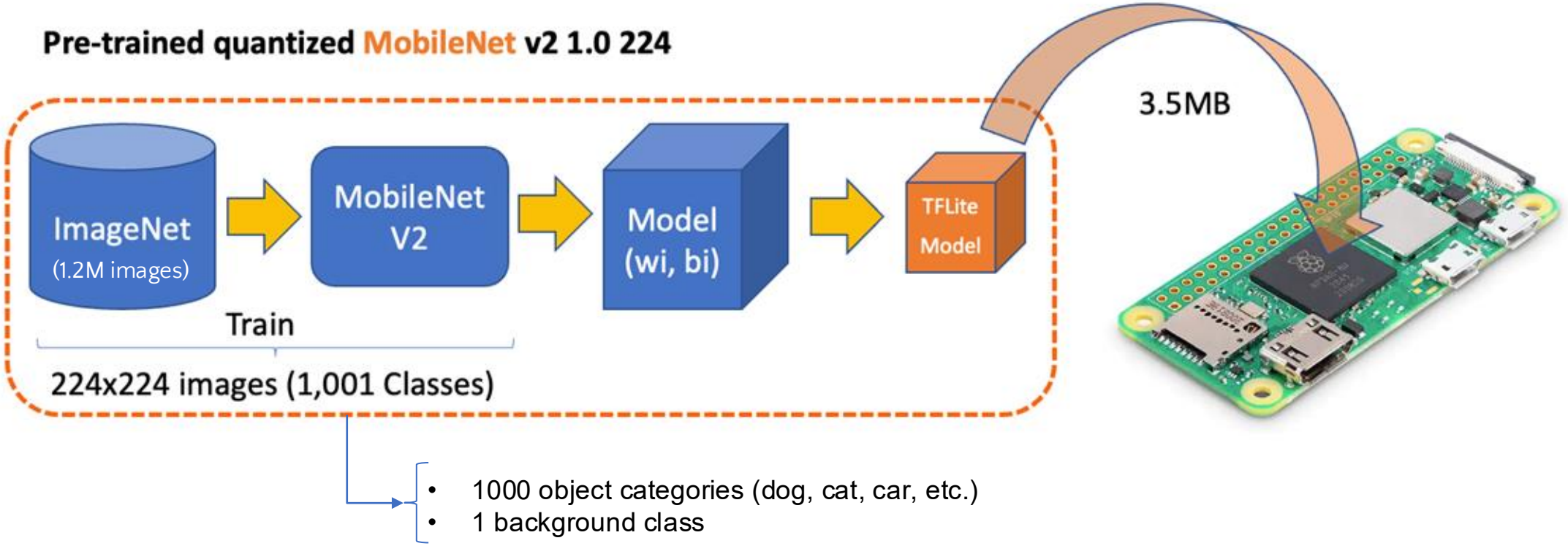
Input Image

Pre-Process

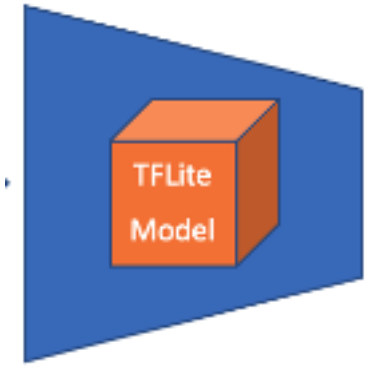
Classification Model

Predict Label

The MobileNet V2 model



TFLite Interpreter




```
input_details
```

```
[{'name': 'input',
  'index': 171,
  'shape': array([  1, 224, 224,   3], dtype=int32),
  'shape_signature': array([  1, 224, 224,   3], dtype=int32),
  'dtype': numpy.uint8,
  'quantization': (0.0078125, 128),
  'quantization_parameters': {'scales': array([0.0078125], dtype=float32),
   'zero_points': array([128], dtype=int32),
   'quantized_dimension': 0},
  'sparsity_parameters': {}}]
```

Input Image Shape Signature

Pixel data type

Quantization Parameters

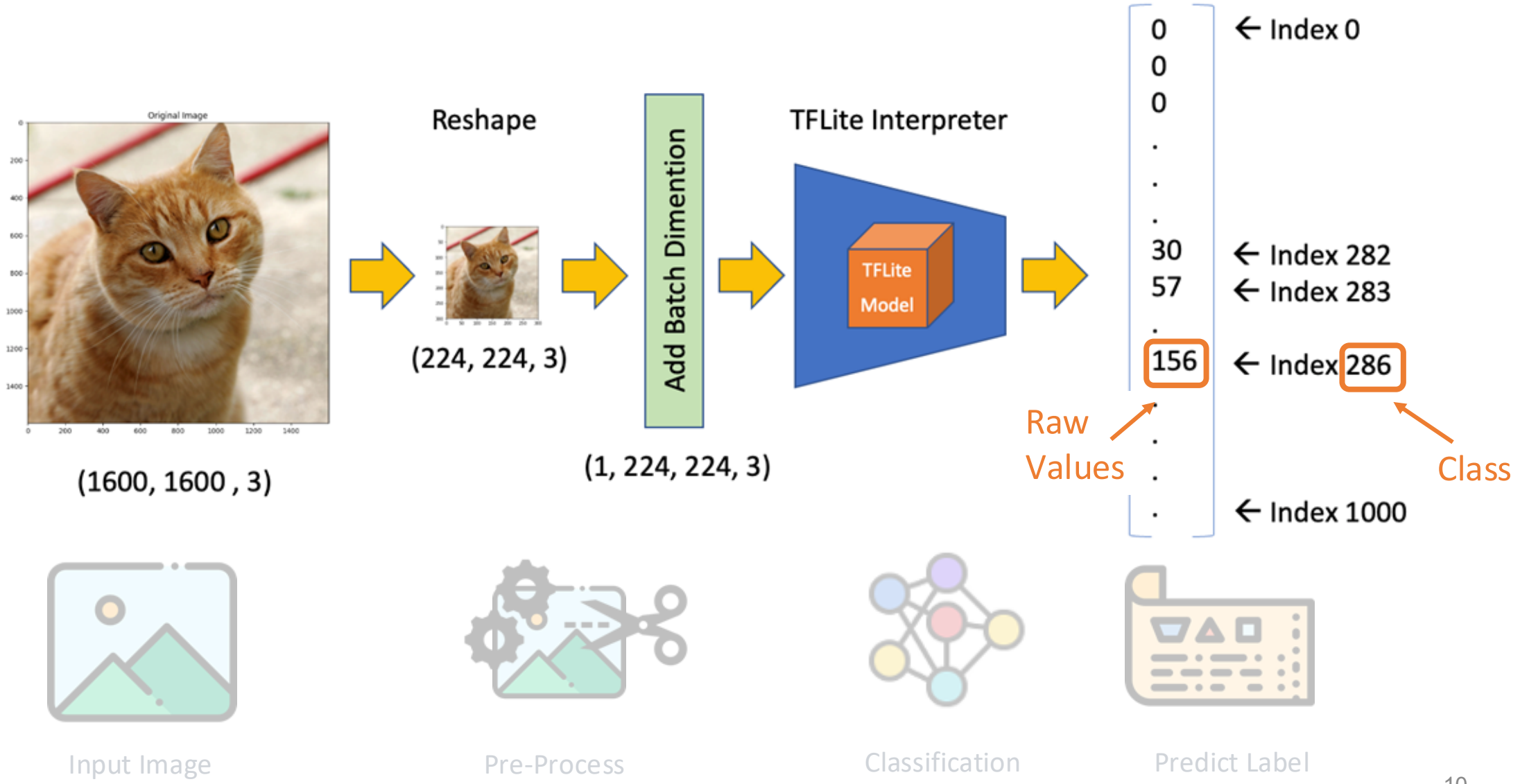
```
output_details
```

```
[{'name': 'output',
  'index': 172,
  'shape': array([   1, 1001], dtype=int32),
  'shape_signature': array([   1, 1001], dtype=int32),
  'dtype': numpy.uint8,
  'quantization': (0.09889253973960876, 58),
  'quantization_parameters': {'scales': array([0.09889254], dtype=float32),
   'zero_points': array([58], dtype=int32),
   'quantized_dimension': 0},
  'sparsity_parameters': {}}]
```

Output model

Quant Dim 0: One set of parameters for the entire output tensor

Making inferences with MobileNet V2



LiteRT Runtime

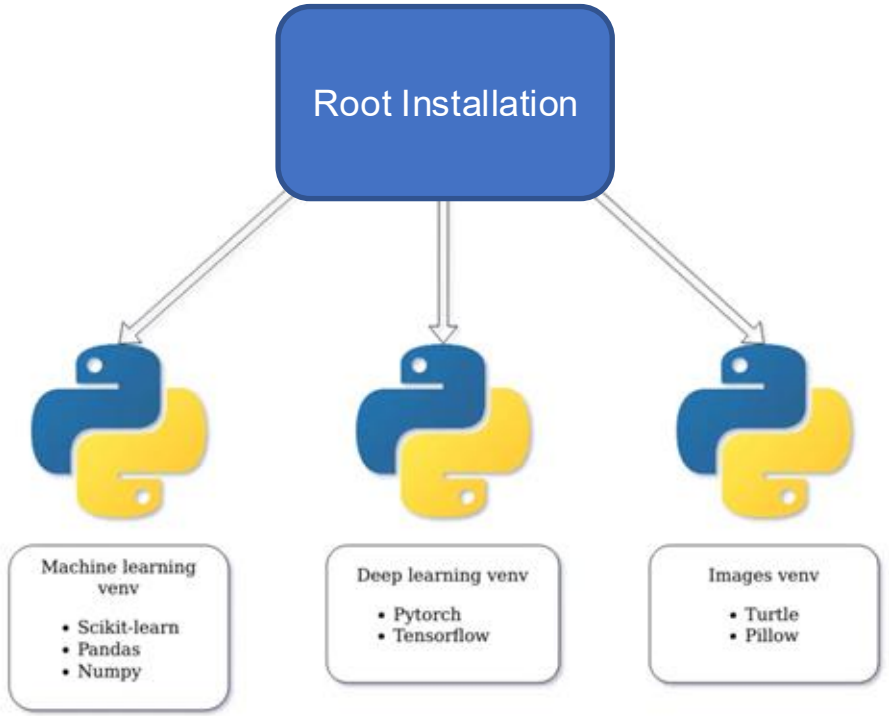
Setting up a Virtual Environment

Activate the environment:

```
source ~/tflite_env/bin/activate
```

To exit the virtual environment, use:

```
deactivate
```



LiteRT Setup

What is LiteRT?

Google's on-device framework for high-performance ML & GenAI deployment on edge platforms, via efficient conversion, runtime, and optimization.

We'll use the LiteRT runtime, an advanced GPU/NPU acceleration which delivers superior ML & GenAI performance, making on-device ML inference easier than ever.

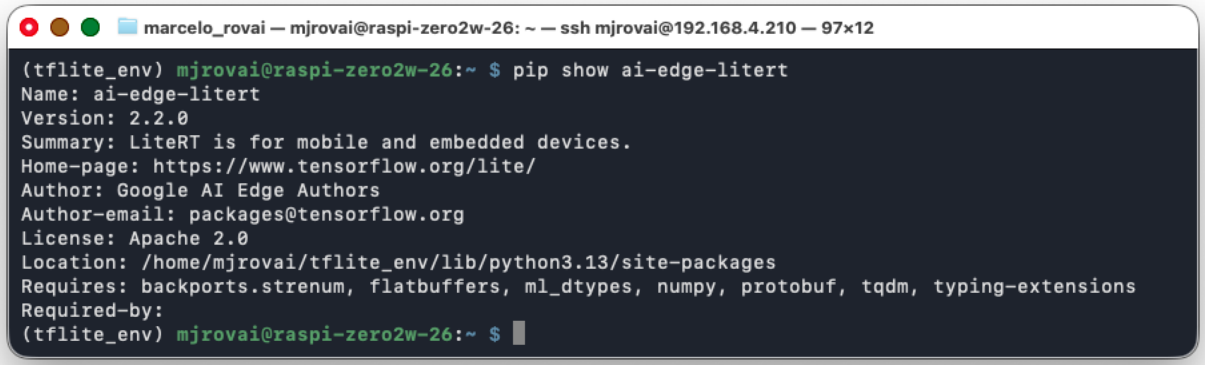
```
pip install ai-edge-litert
```

```
Successfully installed ai-edge-litert-2.2.0 backports.strenum-1.2.8 flatbuffers-25.12.19 ml_dtypes-0.6.0 protobuf-7.35.1
```

```
sudo reboot
```

Verify installation:

```
pip show ai-edge-litert
```



Creating a working directory & get the model

```
Documents/
├── TFLITE/
│   └── IMG_CLASS/
│       ├── models/
│       │   ├── mobilenet_v2.tflite
│       │   └── labels.txt
│       └── images/
│           └── test_image.jpeg
```

You only need the .tflite and the labels.txt files.
You can delete the other files in the models directory.

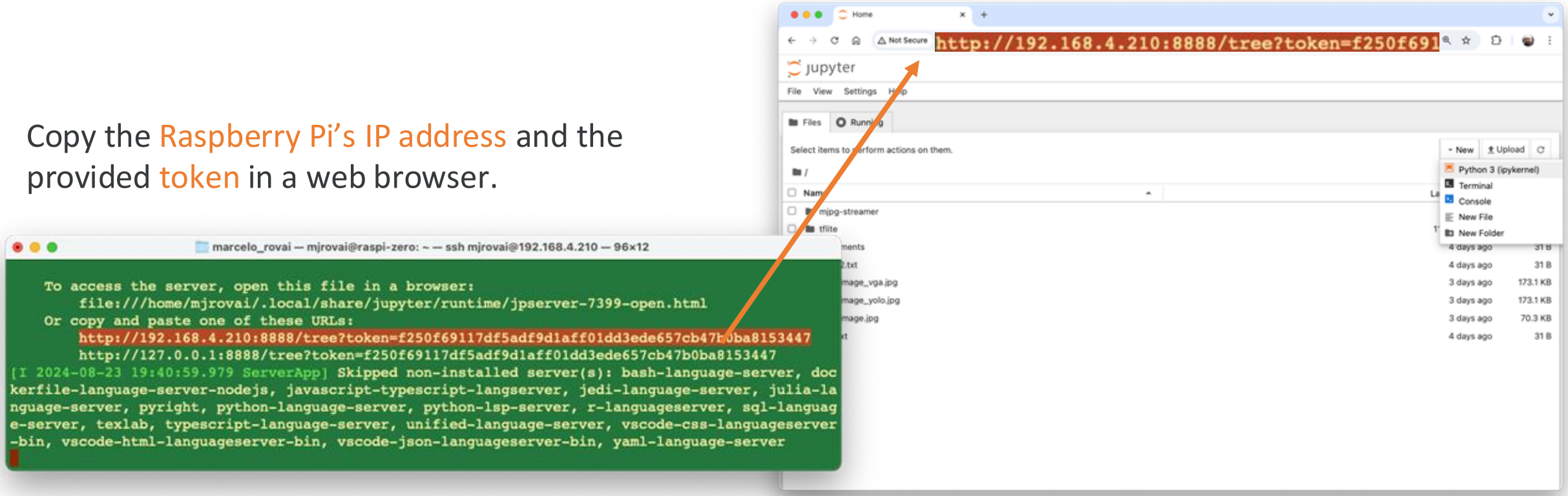
Go to the models/ folder and get the Model and labels

```
wget https://storage.googleapis.com/download.tensorflow.org/models/tflite_11_05_08/mobilenet_v2_1.0_224_quant.tgz

tar xzf mobilenet_v2_1.0_224_quant.tgz

wget https://raw.githubusercontent.com/Mjrovai/EdgeML-with-Raspberry-Pi/refs/heads/main/IMG_CLASS/models/labels.txt
```

Running up Jupyter Notebook

jupyter notebook --ip=[YOUR IP ADDREES] --no-browser

Copy the Raspberry Pi's IP address and the provided token in a web browser.



Verifying the Setup


```
from ai_edge_litert.interpreter import Interpreter
import numpy as np
from PIL import Image

print("NumPy:", np.__version__)
print("Pillow:", Image.__version__)

# Try to create a TFLite Interpreter
model_path = "./models/mobilenet_v2_1.0_224_quant.tflite"
interpreter = tflite.Interpreter(model_path=model_path)
interpreter.allocate_tensors()
print("TFLite Interpreter created successfully!")
```

Note that on video, the runtime can show the old line:
import tflite_runtime.interpreter as tflite

Raspberry Pi Inference:

10_Image_Classification.ipynb

Making inferences: Static Images and Camera

[PREDICTION]	[Prob]
tiger cat	: 37%
Egyptian cat	: 27%
tabby	: 16%
lynx	: 10%
carton	: 2%

[PREDICTION]	[Prob]
coffee mug	: 99%
cup	: 0%
whiskey jug	: 0%
teapot	: 0%
water jug	: 0%

Questions?

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